Regal Meeting — Mobile App API Documentation

Version: 2.0
Last updated: 2026-08-03
Audience: AI agents and engineers maintaining the Regal Meeting mobile companion (regal-meeting-mobile, Expo / React Native).

This document is the source of truth for syncing the mobile app with the web app at https://meet.regalmesh.com. Both clients share one Supabase backend (Regal Mail project).

1. Project Identity & Endpoints

Item	Value
Product name	Regal Meeting
Company	Spatial Regal Digital Ltd
Web app	https://meet.regalmesh.com
Deep link scheme	regalmeet://
Universal links	https://meet.regalmesh.com/meeting/{CODE}
Supabase project ref	xexnwcmqnelgzuqhkvtx
Supabase URL	https://xexnwcmqnelgzuqhkvtx.supabase.co
Edge Functions base	https://xexnwcmqnelgzuqhkvtx.supabase.co/functions/v1

> ' Use the publishable (anon) key from the Supabase dashboard. Never embed the serv
> ' Auth and Meeting data live in the same project as Regal Mail (@regalmail.me).

Mobile env (EXPO_PUBLIC_*)

Variable	Purpose
EXPO_PUBLIC_SUPABASE_URL	Shared Supabase URL
EXPO_PUBLIC_SUPABASE_ANON_KEY	Publishable key
EXPO_PUBLIC_MEET_URL	https://meet.regalmesh.com
EXPO_PUBLIC_R2_PUBLIC_URL	Optional R2 CDN
EXPO_PUBLIC_TURN_URLS	Comma-separated TURN URLs (parity with web VITE_TURN_*)
EXPO_PUBLIC_TURN_USERNAME / _CREDENTIAL	TURN credentials

2. Authentication

2.1 Email / password

Standard Supabase Auth (signUp / signInWithPassword / signOut / password reset).
Mobile redirect: regalmeet://auth-callback.

2.2 Regal Mail SSO

Accounts on @regalmail.me use the same Supabase project — no bridge needed when unified.

- Password: signInWithPassword with a normalized @regalmail.me address
- Magic link: signInWithOtp !' regalmeet://auth-callback?provider=regal-mail

Web uses the same rules (regal-mail-bridge only if Meeting/Mail projects diverge).

3. Plans & billing

Edge Function: meeting-billing

Plan IDs: free | pro | vault_plus | ultra | business_education | business_corporate | team

Plan	Max participants	Duration	SFU	Recording
free	110	60 min	no	no

pro	200	8 h	yes	yes
business_education	300	8 h	yes	yes
vault_plus / ultra / team / business_corporate	500	up to 24 h	yes	yes

Upgrade UX: open https://regalmesh.com/dashboard?upgrade=pro (Paystack on Regal Cloud).

Enforce on mobile: host plan limits duration, screen share, recording, and SFU eligibility (same as web).

4. Meetings

Tables: meetings, meeting_participants, scheduled_meetings, user_recent_meetings, …
Lookup RPC / helper: get_meeting_by_code / fetchMeetingByCode.

Invite link format:

https://meet.regalmesh.com/meeting/{CODE}
regalmeet://meeting/{CODE}

Encrypted mobile-only P2P rooms: codes starting with CALL- (web must not initiate these).

5. Lobby

Realtime channel: lobby-{meetingCode}
Events: knock, admit, deny — same payload shape as web.

6. WebRTC signalling (web ↔ mobile)

Channel: meeting-{meetingCode}

Interop rule

Clients must dual-emit web + mobile events via buildOutboundBroadcasts / normalizeInboundSignal (webrtcSignaling.ts):

Logical type	Web event	Mobile legacy events
join / rejoin	signaling {type:'rejoin'}	peer-join
leave	signaling {type:'leave'}	peer-leave
offer / answer	signaling + typed data	offer / answer
ICE	signaling {type:'ice-candidate'}	ice and ice-candidate

ICE servers: STUN (Google + Cloudflare + Twilio) + project TURN; fallback openrelay TURN when unset (iceServers.ts).

7. Media topology

Threshold: 12 participants (SFU_AUTO_THRESHOLD).

Mode	Behavior
mesh	Full P2P ("d12)
sfu	Cloudflare Realtime SFU via Edge Function meeting-sfu
host-hub	Fallback when SFU unavailable / plan lacks SFU

Topology channel: meeting-topology-{meetingId} event topology.
SFU registry: meeting-sfu-registry-{meetingId} events tracks-published, request-tracks.

Mobile ports: lib/cloudflareSfu.ts, hooks/useCloudflareSfu.ts, hooks/useMeetingTopology.ts.

8. File sharing & R2

Edge Functions: meeting-r2 (presign), cloudflare-health.
Folders: avatars, meeting-files. Same pipeline on web and mobile.

9. Chat, hands, reactions, presentation, pin

Feature	Channel	Event(s)
Chat	meeting-chat-{code}	message
Hands	meeting-hands-{code}	hand-raised
Reactions	meeting-reactions-{code}	reaction broadcast
Presentation	present channel	start / stop present
Pin	pin channel	pin target

10. Captions

Table: meeting_captions
Realtime: postgres_changes INSERT on meeting_id.

- Web: Web Speech API producer + display
- Mobile: display + typed caption producer (shared DB so both platforms see the same lines)

11. Recordings

Table: meeting_recordings
Host-only; gated by plan recording.

- Web: canvas composite + MediaRecorder !' upload
- Mobile: local audio capture (expo-av) !' R2 upload + same row status lifecycle

12. Mobile-only features

Keep on mobile (do not require on web):

- Regal Number dialer (user_phone_numbers)
- Encrypted P2P CALL-* rooms / app_calls
- Contacts, invite ring, biometrics, Expo push, background call handling

Edge Functions (mobile): initiate-app-call, notify-meeting-ring.

13. Profiles, settings, notifications

Aligned with web tables: profiles, user_settings, notifications / push tokens.
Activity logging helper shared conceptually with web logActivity.

14. Branding

Item	Value
Domain	meet.regalmesh.com / regalmesh.com
Company	Spatial Regal
Accent	#ff5c28 (primary orange)
Dark chrome	#0c0e15 splash / backgrounds

15. Parity checklist (mobile sync)

- [x] Domain / deep links !' meet.regalmesh.com
- [x] Dual signalling with web
- [x] Shared ICE / TURN config
- [x] SFU + topology

- [x] Plan limits + Regal One panel
- [x] Regal Mail sign-in
- [x] Captions subscribe + publish
- [x] Recording (host, plan-gated)
- [x] R2 file/avatar uploads
- [x] Spatial Regal branding

16. Recommended SDKs

Platform	Packages
React Native (Expo)	@supabase/supabase-js, react-native-webrtc, react-native-url-polyfill, Expo Router

Init:

ts
```
import { createClient } from '@supabase/supabase-js';
export const supabase = createClient(
process.env.EXPO_PUBLIC_SUPABASE_URL!,
process.env.EXPO_PUBLIC_SUPABASE_ANON_KEY!,
{ auth: { persistSession: true, autoRefreshToken: true } }
);
```